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SYSTEM AND METHOD FOR TRANSAORTIC DELIVERY OF A PROSTHETIC HEART VALVE

Abstract

A delivery system and method for delivering a prosthetic heart valve to the aortic valve annulus via a transaortic approach. The system includes a balloon catheter for delivering a balloon-expandable prosthetic heart valve through an introducer from an approach from outside the patient, through a minimally-invasive opening in the chest cavity through an incision in the aorta, and into the aorta and aortic annulus. The balloon catheter includes a flexible, tapered tip just distal of the balloon to facilitate positioning of the prosthetic heart valve in the proper orientation within the aortic annulus. The prosthetic heart valve may be installed over the existing calcified leaflets, and a pre-dilation valvuloplasty procedure may also be utilized.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/544,266, filed Dec. 18, 2023, which is a continuation of U.S. patent application Ser. No. 16/178,537, now U.S. Pat. No. 11,896,485, filed Nov. 1, 2018, which is a divisional of U.S. patent application Ser. No. 14/215,399, now U.S. Ser. No. 10/149,757, filed Mar. 17, 2014, which claims priority to U.S. Provisional Patent Application No. 61/794,889, filed Mar. 15, 2013, all of which are incorporated by reference herein in their entireties.

FIELD OF THE INVENTION

[0002] The present invention relates to methods and systems used to deliver a prosthetic valve to a heart. More specifically, the present invention relates to methods and apparatus for transaortically replacing a heart valve without opening the chest cavity and with or without placing the patient on bypass, the latter being termed “off-pump.”

BACKGROUND OF THE INVENTION

[0003] Heart valve replacement may be indicated when there is a narrowing of the native heart valve, commonly referred to as stenosis, or when the native valve leaks or regurgitates, such as when the leaflets are calcified. When replacing the valve, the native valve may be excised and replaced with either a biologic or a mechanical valve.

[0004] Conventional heart valve surgery is an open-heart procedure conducted under general anesthesia. An incision is made through the patient's sternum (sternotomy), and the patient's heart is stopped while blood flow is rerouted through a heart-lung “cardiopulmonary” bypass machine. In recent years, advancements in “minimally-invasive” surgery and interventional cardiology have encouraged some investigators to pursue percutaneous replacement of the aortic heart valve. Minimally invasive heart valves are known using expandable stents as anchors for a valve structure. Edwards Lifesciences has developed a balloon-expandable stent integrated with a bioprosthetic valve, which is known as the SAPIEN® valve. The stent/valve device is deployed across the native diseased valve to permanently hold the valve open, thereby alleviating a need to excise the native valve. The SAPIEN® device may be delivered in a cardiac catheterization laboratory under local anesthesia using fluoroscopic guidance, thereby avoiding general anesthesia and open-heart surgery.

[0005] U.S. Patent Publication No. 2002/0151970 to Garrison et al. describes a two-piece device for replacement of the aortic valve that is adapted for delivery through a patient's aorta. A stent is endovascularly placed across the native valve, then a replacement valve is positioned within the lumen of the stent and connected thereto. By separating the stent and the valve during delivery, a so-called “two-stage” approach, the profile of the delivery system can be reduced. Both the stent and a frame of the replacement valve may be balloon- or self-expandable.

[0006] Some researchers propose implanting prosthetic heart valves at the aortic annulus through a ventricular approach. For instance, Christoph H. Huber of the Brigham and Women's Hospital of Harvard Medical School, and others, have proposed a procedure in which a self-expanding valve stent is implanted at the aortic position using a direct-access transapical approach. (E.g., Huber, et

al. Direct-access valve replacement a novel approach for off-pump valve implantation using valved stents. J Am Coll Cardiol 2005; 46:366-70). More recently, Bergheim in U.S. Patent Publication No. 2005/0240200 discloses another transapical approach in which either a balloon- or self-expanding valve may be implanted, and also proposes removing or decalcifying stenotic valves. U.S. patent application Ser. No. 11/280,063, filed Nov. 16, 2005, the entire contents of which are expressly incorporated herein by reference, discloses a transapical heart valve delivery system and method, where an expandable prosthetic heart valve is delivered to a desired position via an incision in the apex of the heart.

[0007] The selection of the best method for replacing a heart valve is highly dependent upon the condition of a particular patient. For some patients, traditional open heart surgery can be the preferred option. For other patients, such as those who are in extremely frail condition, a minimally-invasive implantation may be desired. It is important for the treating physician to have multiple options for prosthetic heart valve replacement, in order to best serve the needs of a particular patient.

[0008] There is a need for additional and improved methods and apparatuses for replacing a heart valve without open-heart surgery or cardiopulmonary bypass. The current invention satisfies this need.

SUMMARY OF THE INVENTION

[0009] Preferred embodiments of the present invention provide a heart valve delivery system for delivery of a prosthetic (i.e., replacement) heart valve to a native valve site without an open chest procedure. Methods according to the invention may include a transaortic approach, wherein an incision is made in the aorta through which a valve delivery catheter is advanced. The delivery system includes a valve delivery catheter having a distal tapered tip, which may have a curved portion, to facilitate advancement and positioning of the valve in a native valve, such as a calcified aortic valve.

[0010] In accordance with one aspect, the present invention provides an off-pump, minimally-invasive surgical method of implanting a prosthetic heart valve to an aortic valve annulus of a patient, which may be performed while the patient's heart remains beating. The method includes providing a balloon-expandable prosthetic heart valve mounted over a balloon on a distal end of a balloon catheter. The physician creates a small puncture in the chest wall, and then creates a puncture through the wall of the aorta at or near the aortic arch. The physician then advances an introducer sheath through the respective punctures (i.e., through the chest wall and the aortic wall) and into a position above the aortic valve. A delivery catheter, with the prosthetic valve on its distal end, is passed through the introducer sheath into the ascending aorta. The distal end of the delivery catheter is advanced to cross the aortic valve, with the tapered tip providing a smooth transition through the native valve leaflets so that the prosthetic heart valve is positioned within the aortic annulus. The prosthetic heart valve is expanded (e.g., by expanding a balloon on the delivery catheter) at the aortic annulus.

[0011] The method may also include sealing the aorta against the introducer sheath, which may be performed using purse string sutures placed in the aortic tissue around the introducer sheath to cinch around the introducer sheath. This allows a hemostatic seal when the purse strings are tightened around the sheath. The purse strings help with adding tension to the artery walls as it is being brought around the introducer sheath body.

[0012] The balloon catheter may incorporate a steering mechanism, and the method may further include steering the balloon catheter within the aortic arch to facilitate positioning the prosthetic heart valve within the aortic annulus.

[0013] The balloon catheter may include a tapered segment located just distal to the balloon starts from a tip (which may be narrow but blunted to prevent damage to tissue) and widens to the diameter of the balloon with prosthetic heart valve crimped thereon. The tip may be bent to form a curve which can facilitate advancement of the catheter into the native aortic valve annulus, which

may be performed without the need for a guidewire. The curved tip also may serve a protective role, and can allow the catheter to be positioned away from the ventricular walls of the heart.

[0014] Note that a guidewire may or may not be used during various portions, or even during the entirety of, the procedures disclosed herein.

[0015] A method of the invention may further involve leaving the native aortic valve leaflets in place such that inflating the balloon expands the prosthetic heart valve into contact therewith. Furthermore, a pre-dilation balloon catheter may be inserted prior to the valve delivery balloon catheter, and a balloon thereon inflated to pre-dilate the aortic annulus. Alternatively, the method may include expanding the prosthetic heart valve into contact with a prosthetic heart valve that was previously implanted at the aortic annulus

[0016] Another off-pump, minimally-invasive surgical method of delivering a prosthetic heart valve to an aortic valve annulus of a patient while the patient's heart remains beating includes providing an expandable prosthetic heart valve in an expanded state and a delivery catheter having a distal end. The heart valve is crimped to a contracted state over the delivery catheter distal end. A relatively small access opening is made in the chest wall to access the aorta. For example, an intercostal access opening (e.g., in the right second intercostal space), also known as a mini-thoracotomy (e.g., right anterior) may be created to access the aorta of the patient, or a ministernotomy may be performed to provide such access, or a subclavian puncture. With access provided to the aorta, an aortic puncture can be formed at or near the aortic arch. An introducer sheath is inserted into the patient's chest via the chest access opening and through the aortic puncture, and the delivery catheter can be advanced through the introducer sheath and into a position in the ascending aorta adjacent the aortic valve annulus. The distal end of the delivery catheter is advanced and steered into the aortic annulus so that the prosthetic heart valve is properly positioned and oriented within the aortic annulus. Finally, the prosthetic heart valve expands at the aortic annulus into contact therewith.

[0017] The delivery catheter may include a balloon or other expandable member on its distal end, and the prosthetic heart valve may include a balloon-expandable stent, wherein the step of expanding includes expanding the balloon (e.g., by injecting fluid into the balloon) or other expandable member to expand the prosthetic heart valve outward into contact with the aortic annulus. The delivery catheter may include a restraining sheath or other restraining device on its distal end which holds the prosthetic heart valve in proper position on the delivery catheter. The prosthetic heart valve may include a self-expanding stent, wherein the step of expanding includes releasing the prosthetic heart valve from a restraining device (e.g., restraining sheath) to permit the self-expanding prosthetic heart valve to expand outward into contact with the aortic annulus. For balloon expandable or self-expanding prosthetic heart valve, the native aortic valve leaflets (or leaflets or a previously-implanted prosthetic valve) may be left in place such that the prosthetic heart valve expands into contact therewith. Prior to expanding the expandable prosthetic heart valve, a pre-dilation balloon may be advanced through the puncture and inflated to pre-dilate the aortic annulus.

[0018] The present invention further encompasses a minimally-invasive prosthetic heart valve delivery system, including an introducer sheath having a lumen therethrough of no greater than 24 French and a balloon catheter having a balloon on a distal end, the balloon catheter further including a tapered tip for improved advancement into the aortic annulus. In one embodiment, the tapered tip tapers from a maximum width of 4 mm to 10 mm to a minimum width (at the distal tip) of 2 mm or less. In a further embodiment, the distal tip has a diameter of about 1 mm. The tapered tip may have a length of 10 mm to 40 mm between the maximum width and the distal tip, and the tapering may be consistent along the length thereof. The system may include a balloon-expandable prosthetic heart valve crimped over the balloon, wherein the outer dimension of the balloon catheter with the prosthetic heart valve crimped thereon is small enough to pass through the introducer sheath lumen, and the tapered tip provides a generally smooth transition along the

delivery catheter from the tip to the crimped prosthetic valve.

[0019] The system may be relatively short, such that the balloon catheter has a working length sized to fit into the introducer of no more than about 24 inches (61 cm). At the same time, the introducer sheath may have a total length of no more than about 13 inches (33 cm).

[0020] A further understanding of the nature and advantages of the present invention are set forth in the following description and claims, particularly when considered in conjunction with the accompanying drawings in which like parts bear like reference numerals.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0021] Features and advantages of the present invention will become appreciated as the same become better understood with reference to the specification, claims, and appended drawings wherein:

[0022] FIG. 1A is a schematic frontal view of a patient showing the location of an intercostal incision and aortic incision providing access into the aortic arch of a patient in accordance with an embodiment of the current invention;

[0023] FIG. 1B is a cross-sectional view through the left side of a patient's heart showing a procedure for advancing a guidewire and introducer sheath into the patient in accordance with an embodiment of the current invention.

[0024] FIGS. 2A-2B are cross-sectional views through the left side of a patient's heart showing a procedure for dilating a calcified aortic annulus prior to implantation of a prosthetic heart valve in accordance with an embodiment of the current invention;

[0025] FIGS. 3A-3E are cross-sectional views through the left side of a patient's heart showing several steps in a procedure for implanting a prosthetic heart valve in accordance an embodiment of the current invention;

[0026] FIG. 4 is a side elevational view of an introducer in accordance with an embodiment of the current invention;

[0027] FIG. 5 is a perspective view of a valve delivery catheter in accordance with an embodiment of the current invention;

[0028] FIG. 6 is a side view, in partial cross-section, of a catheter distal portion in accordance with an embodiment of the current invention;

[0029] FIG. 7 is a side view, in partial cross-section, of a catheter distal portion in accordance with an embodiment of the current invention;

[0030] FIG. 8 is a side view, in partial cross-section, of a catheter distal portion in accordance with an embodiment of the current invention;

[0031] FIGS. 9A-9B are a side views, in partial cross-section, of a catheter distal portion in accordance with an embodiment of the current invention; and

[0032] FIG. 10 is a side view, in partial cross-section, of a catheter distal portion in accordance with an embodiment of the current invention.

[0033] FIG. 11 is side view, in partial cross-section, of another embodiment of a tapered distal tip.

DETAILED DESCRIPTION OF THE INVENTION

[0034] The heart is a hollow muscular organ of a somewhat conical form. It lies between the lungs in the middle mediastinum and is enclosed in the pericardium. The heart rests obliquely in the chest behind the body of the sternum and adjoining parts of the rib cartilages, and projects farther into the left than into the right half of the thoracic cavity so that about one-third is situated on the right and two-thirds on the left of the median plane. The heart is subdivided by septa into right and left halves, and a constriction subdivides each half of the organ into two cavities, the upper cavity being called the atrium, the lower the ventricle. The heart therefore consists of four chambers; the right

and left atria, and right and left ventricles.

[0035] As seen in FIG. 1A, the aorta **10** extends above the heart **12**, starting at the aortic valve **14** and forming the aortic arch **15**, which extends above and to the left of the heart (from the perspective of the patient). Access to the aorta **10** and aortic valve **14** may be made via several approaches. For many patients, a direct line-of-sight approach to the aortic valve annulus **16** can be attained, via the aorta **10**, starting at a thoracic incision **18** in the patient **20** as shown in dashed line, which in the particular embodiment depicted is positioned at the second right intercostal space. Such an approach is often termed a “mini-thoracotomy.” Once access to the aorta **10** is provided (e.g., via intercostal incision), an aortic incision **22** is made in the aorta **10**. As depicted in FIG. 1B, a guidewire **23** may be introduced into the aorta **10** via the thoracic incision **18** and aortic incision **22**. The guidewire **23** may be advanced so that its distal end **25** crosses the aortic valve **14** and enters the left ventricle **17**. Note that the use of the guidewire **23** is optional, depending on the particular application. An introducer sheath **24** may be advanced into the aorta **10** via the thoracic incision **18** and aortic incision **22** (and over the guidewire **23**, if present), as depicted in FIG. 1B. The introducer sheath **24** may be flexible or substantially rigid, and may be straight and/or curved. The introducer sheath **24** diameter may be no greater than 24 French, and 22 or 24 Fr may be preferred, or even smaller, such as 18 F or smaller, depending on crimped valve/balloon outer diameter limitations. In one embodiment, the introducer sheath **24** is substantially rigid and substantially straight, and can be advanced in a generally straight line from the thoracic incision **18** thru the aortic incision **22** and adjacent the aortic valve annulus **16**. The physician may locate the distal opening **26** of the introducer sheath **24** just to the outflow side of the aortic valve, as seen in FIG. 1B. The position of the introducer sheath **24** relative to the aortic valve **14**, as well as the position of other elements of the system, may be monitored using radiopaque markers and fluoroscopy, and/or using other imaging systems such as transesophageal echo, transthoracic echo, intravascular ultrasound imaging (IVUS), and/or an injectable dye that is radiopaque. The guidewire may be advanced into the heart prior to introduction of the introducer sheath, with the sheath being advanced over the guidewire. The order may also be reversed, with the introducer sheath advanced first, and the guidewire introduced through the introducer sheath.

[0036] Methods according to the invention may include using an introducer sheath but without the use of a guidewire, using a guidewire without the use of an introducer sheath, using both a guidewire and introducer sheath (as depicted in FIG. 1B), or using neither a guidewire nor introducer sheath.

[0037] In one embodiment, sutures **28** are passed through the aortic tissue and around the aortic incision **22**, such as in a so-called purse-string configuration, to hold the aortic tissue tightly closed (e.g., against the introducer sheath **24**) to prevent blood from leaking out of the aorta. The sutures **28** may be passed through the aortic tissue prior to, during, or after formation of the aortic incision **22**, with tension applied to the sutures **28** to hold the aortic incision **22** closed and/or hold the aortic tissue tightly against the introducer sheath **24**.

[0038] In one embodiment of the present invention, a surgeon implants a prosthetic heart valve over the existing native leaflets, which may be calcified. There are procedures and devices for removing calcified leaflets, but such removal may not be necessary. Therefore, a heart valve replacement procedure that installs the prosthetic heart valve directly over (and thereby containing) the native leaflets may be preferred. However, the calcified native leaflets may present issues in advancing devices through the native valve annulus.

[0039] Those skilled in the art will recognize that it may be necessary to pre-dilate the leaflets and annulus of the stenotic aortic valve before deploying a prosthetic valve within the aortic valve. FIGS. 2A and 2B depict a valvuloplasty procedure that may be initially performed to compress the native aortic heart valve leaflets **30** outward against the aortic valve annulus **16** and/or aorta **10**. As mentioned above, the native aortic valve leaflets **30** may be substantially calcified, and the valvuloplasty may be necessary to crack and otherwise force apart hardened tissue. Pre-dilatation

may increase the flow area through the aortic valve **14** and create an opening in the native leaflets **30** of sufficient size to receive the new prosthetic valve. Pre-dilatation may be achieved using an expandable member, such as a dilatation catheter **32** having a dilatation balloon **34** thereon. One example of pre-dilatation of a valve annulus is seen in U.S. Pat. No. 6,908,481 to Cribier, issued Jun. 21, 2005 and expressly incorporated by reference herein.

[0040] FIG. 2A illustrates a dilatation catheter **32** with dilatation balloon **34** at a distal end **36** passed through the introducer sheath **24** and into the aorta **10**, with the distal tip **36** of the dilatation catheter **32** advanced through the aortic valve **14** and into the left ventricle **17** until the dilatation balloon **34** is positioned within the aortic valve annulus **16**. The dilatation catheter distal tip **36** may be tapered and/or curved to facilitate the advancement of the dilatation catheter past the aortic valve leaflets **30** and through the annulus **16**. The dilatation balloon catheter tip **36** may be formed in a similar fashion to the delivery catheter tips discussed below and depicted with respect to FIGS. 6-10 (see element **84** therein).

[0041] As indicated in FIG. 2B, the dilatation balloon **34** is expanded radially outward into contact with the native aortic valve leaflets **30**. The particular dilatation balloon **34** used in a particular procedure and patient may be selected, based on size and other characteristics (e.g., of the native aortic valve), so that it expands outward and nominally compresses the native valve leaflets **30** against the surrounding annulus **16** and/or aortic walls. Assessing the characteristics (size, calcification, etc.) of the particular patient's native aortic valve may be performed using various known methods and devices, including ultrasound. As a result of the valvuloplasty procedure depicted in FIG. 2B, the native valve leaflets are compressed outwardly against the aortic wall and a substantially circular orifice may result. Additional details regarding pre-dilatation and valve replacement can be found in U.S. Patent Publication No. 2003/0014104, filed May 2, 2002, the contents of which are expressly incorporated by reference herein.

[0042] The devices and methods disclosed herein are particularly well-suited for replacing a stenotic aortic valve, and as such the native valve dilatation procedure depicted in FIGS. 2A-2B may precede the valve implantation so as to smooth out the contours of the annulus **16** and leaflets **30**. It should be noted, however, that the procedure described herein may be performed without valve pre-dilation. Furthermore, the present procedure may be performed as a first time valve implant or to supplement a previous implant. A relatively large proportion of recipients of prosthetic heart valves are older, typically older than 60. Over time, prosthetic heart valves have been known to show reduced performance and even failure. Re-operating on elderly and/or frail patients may be problematic. However, a minimally-invasive procedure such as disclosed herein can eliminate open-heart surgery and potentially cardiopulmonary bypass, and may be desirable for an aging patient. Therefore, the present invention contemplates transaortic implantation of a prosthetic heart valve within an existing prosthetic valve implant. In such a case, a pre-dilation step is typically not necessary, though it is conceivable.

[0043] The prosthetic heart valve implantation procedure described herein may be performed in conjunction with cardiopulmonary bypass, or without such bypass in a so-called off-pump procedure. The necessity for bypass depends on a number of factors, including the patient's overall health, age, vulnerability to such a procedure, and condition of the native valve. The implantation procedure may preferably be performed off-pump.

[0044] FIGS. 3A-3E depict a method of deploying and implanting a prosthetic heart valve **40** with a delivery catheter **38** with tapered tip **42** in a transaortic approach according to an embodiment of the present invention. The physician may first size the native aortic valve (or previously-implanted prosthetic valve) using, e.g., a physical sizer, echocardiography, and/or other known method (or equivalent thereof). Note that the expandable prosthetic valve **40** may be provided pre-crimped onto the balloon **44** of the delivery catheter **38**, or the physician or other operating room personnel may crimp the expandable prosthetic valve **40** over the balloon **44** of the delivery catheter **38**. With the expandable prosthetic valve **40** positioned thereon, the physician advances the delivery catheter

38 through the previously-positioned introducer sheath **24**.

[0045] FIG. 3A shows a delivery catheter introducer sheath **24** positioned in the left ventricle through the aortic incision **22** and over the guidewire **23** that extends downward through the native aortic valve **14**. Note that the delivery catheter introducer sheath **24** may be the same introducer sheath depicted in FIGS. 1B-2B for use with the dilatation catheter **32**, or may be a (separate) sheath dedicated to use with the delivery catheter.

[0046] FIG. 3B shows the advancement of the delivery catheter **38** over the guidewire **23** and through the introducer sheath **24** until the prosthetic heart valve **40** is located at the aortic annulus **16** and between the native aortic leaflets **30**. FIG. 3C also illustrates retraction of the introducer sheath **24** from its more forward position in FIG. 3B. Radiopaque markers may be provided on the distal end of the introducer sheath **24** and/or valve **40** and/or delivery catheter **38** to more accurately determine their relative positions with respect to each other and to the native structures. During advancement of the delivery catheter **38**, a pusher **39** may be positioned just proximal of the crimped prosthetic heart valve **40**, which may prevent the prosthetic heart valve **40** from moving proximally with respect to the delivery catheter **38**. The pusher **39** is typically retracted to a position proximal of the balloon **44** prior to balloon inflation, as depicted in FIG. 3C.

[0047] The precise positioning of the prosthetic heart valve **40** may be accomplished using known methods, such as by locating radiopaque markers on the prosthetic heart valve (e.g., at its distal and proximal ends) and/or on the balloon **44** and/or other part of the delivery catheter **38**. The surgeon may adjust the position of the valve **40** by actuating a steering or deflecting mechanism within the delivery catheter **38**. Furthermore, the rotational orientation of the prosthetic valve **40** can be adjusted relative to the cusps and commissures of the native valve by twisting the delivery catheter **38** from its proximal end and observing specific markers on the valve **40** or delivery catheter **42** under fluoroscopy. One of the coronary ostia **48** opening into a sinus of the ascending aorta **10** is shown, and those of skill in the art will understand that it may be desirable to avoid occluding the coronary ostia with the prosthetic valve **40**. It should also be noted that although the native leaflets **30** of the aortic valve **14** are shown coapting in FIG. 3A, and being flexibly displaced by the delivery catheter **38** in FIGS. 3B and 3C, the native leaflets may actually be in quite different positions, such as being compressed further outward against the aortic annulus from a pre-dilation procedure and/or malformed due to extensive calcification.

[0048] FIG. 3C shows the prosthetic heart valve **40** in its contracted or unexpanded state crimped around the delivery catheter balloon **44**. When the physician is satisfied of the proper positioning and rotational orientation of the valve **40**, the delivery catheter balloon **44** is expanded as seen in FIG. 3D. Proper size measurement of the native aortic valve **14** and annulus **16** prior to deployment enables the physician to select an optimum-sized valve **40** that will expand outward into good contact with the aortic valve annulus **16**. Note that the term "good contact" implies sufficient contact to ensure that the prosthetic heart valve **40** does not migrate after implant.

[0049] A number of devices are available to assist in anchoring the prosthetic valve **40** into the aortic annulus, such as barbs and the like. Examples of such anchors for use with a prosthetic heart valve of the present invention are disclosed in co-pending U.S. patent application Ser. No. 10/883,575, filed Jun. 30, 2004, the disclosure of which is expressly incorporated herein by reference. The prosthetic valve **40** can take a variety of different forms but generally comprises an expandable stent portion that supports a valve structure. The stent portion has sufficient radial strength to hold the valve at the treatment site and resist recoil of the stenotic valve leaflets and other adjacent structures. Additional details regarding examples of balloon expandable valve embodiments for use with the invention can be found in U.S. Pat. Nos. 6,730,118 and 6,893,460, the entire contents of both of which are expressly incorporated herein by reference. The prosthetic heart valve **40** may include sufficient irregularity on its outer surface such that it may be anchored in the aortic annulus without the use of barbs or other tissue piercing structures.

[0050] Once the prosthetic valve **40** is properly implanted, as seen in FIG. 3D, the delivery catheter

balloon **44** is deflated, and the entire delivery system including the delivery catheter **38** is withdrawn over the guidewire **23**. The guidewire **23** is then withdrawn, followed by the introducer sheath **24**. Ultimately, the sealing sutures **28** previously described are cinched tight and tied to close the aortic incision **22**, as seen in FIG. 3E.

[0051] It is important to recognize that the heart valve delivery system of the present invention is particularly well-suited for the retrograde, left ventricular apex, “transaortic,” approach. More particularly, the mini-thoracotomy approach requires relatively short instruments. Therefore, the portion of the introducer sheath **24** that extends into the body may desirably be no more than about 8 inches (20 cm) long, and the length of the delivery catheter **38** (and of the dilatation catheter **32**) that may extend into the introducer sheath **24**, i.e., the “working length,” may desirably be no more than about 24 inches (61 cm). Further specifics on general characteristics of delivery catheters and introducer sheaths which may be used with the current invention are presented in co-pending U.S. patent application Ser. No. 11/280,063 filed Nov. 16, 2005, and in co-pending U.S. patent application Ser. No. 11/152,288 filed Jun. 13, 2005, the contents of each of which is expressly incorporated herein by reference. The present system differs in several aspects that make it more suitable for a transaortic approach, although some features are common.

[0052] FIG. 4 depicts an example of an introducer **50** having a delivery sheath **52** (which can be used as the sheath **24** depicted in FIGS. 2-3) which forms an internal lumen **54** through which other devices (e.g., balloon catheters, etc.) may be introduced into the patient's heart. The delivery sheath **52** is coupled to an introducer housing **56** containing a series of valves which provide a seal to prevent fluid (e.g., blood) leakage through the internal lumen **54**. The valves will prevent leakage when the sheath is free of instruments, and also will provide a tight seal around instruments of various sizes (e.g., diameters) as they are introduced through the introducer. For example, the valves seal around both a guidewire and a balloon catheter, whether introduced together or separately. The delivery sheath **52** can extend into the patient's body and into the heart, with the introducer housing **56** located outside the patient's body. The surgeon or other user can thus remotely introduce instruments into the patient's heart by passing the instruments through the proximal opening **58**, into the internal lumen **54**, and out of the distal opening **60** of the delivery sheath **52** of the internal lumen and In a preferred embodiment, the introducer sheath **52** possesses an external hydrophilic coating and has a length of about 10 cm to 40 cm, depending on the particular patient and application, so that it may extend through the thoracic incision and aortic incision (elements **18** and **22**, respectively, in FIG. 1A), into the aorta, and reach the aortic valve annulus.

[0053] A side port tube **64** extends away from the introducer and terminates in a three-way stopcock **66**. The side port tube **64** enters the lumen **54** at a position distal of the sealing valves in the introducer housing **56**. This permits the user to infuse medicaments or other fluids through the lumen **54** of the introducer **50** even if devices such as the balloon catheter are advanced through the proximal opening **58**.

[0054] FIG. 5 depicts an example of a balloon catheter **70** with a balloon **72** thereon, which comprises an assembly of interrelated components along a catheter shaft **74**. A proximal portion **76** has a luer fitting **77** and handle **78** with one or more controls **80**. A distal end portion has the balloon **72** (on which a prosthetic heart valve can be mounted) and terminates at a distal end **82** in a tapered distal tip **84**. Different embodiments of distal portions for such a balloon catheter are also depicted in FIGS. 6 to 10.

[0055] The balloon catheter of the present invention may incorporate relatively sliding concentric inner and outer balloon inflation tubes that attach to opposite ends of the balloon. A detailed discussion of the structure and function of this concentric tube configuration may be found in U.S. Pat. No. 5,968,068, the disclosure of which is expressly incorporated by reference herein.

[0056] FIG. 6 depicts a distal portion of a balloon catheter **70** according to an embodiment of the invention. An expandable prosthetic heart valve **86** is positioned on the balloon **72**. For a valve

being implanted in the aortic annulus via a retrograde approach such as that depicted in FIGS. 1-3, the prosthetic heart valve **86** is positioned with its inlet end **88** positioned distally (with respect to the balloon catheter **70**) of its outlet end **90**. The prosthetic heart valve **86** comprises a support stent **92** and one-way valve assembly **94** (which may be formed of multiple leaflets). The prosthetic heart valve **86** is crimped or otherwise reduced to a valve delivery diameter **96** (which is typically in the range of 24 F or less) to mate closely with the balloon **72** in a generally deflated state and to reduce the overall system diameter for introduction through the patient's anatomy. The valve delivery diameter **96** is thus close in size to the deflated balloon (maximum) diameter **98** (which is typically in the range of approximately 1.4 mm to 2.5 mm).

[0057] A tapered tip **84** is secured at the distal end of the balloon catheter **70**. The tapered tip terminates in a distal tip **82**, which provides a distal opening for guidewire lumen **100**. The tapered tip **84** may start at a very small initial diameter **102** at its distal tip (which may be on the order of 0.04 inches, i.e., similar in size to the guidewire lumen **100**). Going proximally along the tapered tip, the diameters increases up to an overall diameter **104** which may be about 24 F or less or so, which may be substantially equal or similar in size to the valve delivery diameter **96**. The tapered tip may taper from its widest diameter to its tip along a length **103** of about 10 mm to 40 mm to present an angled surface advantageous for advancement through the anatomy.

[0058] The tapered distal tip **84** may be formed of a substantially flexible and soft material, such as polyethylene (which could be low density), Pebax 35D-50D, etc. As the diameter of the tapered distal tip **84** decreases (as it does toward the distal end thereof), the flexibility increases, so that the distal end thereof will give easily if brought into contact with surrounding tissue. The tapered distal tip **84** can thus reduce trauma to surrounding tissue which might come into the catheter tip when the catheter is advanced into the patient.

[0059] As depicted in FIG. 6, the balloon **72** may extend distally under the proximal portion of the tapered distal tip **84** with a tight seal formed therebetween, so that the balloon material at the distal portion is sandwiched between the tapered distal tip **84** and balloon catheter shaft **74**.

[0060] FIG. 7 depicts a further embodiment, wherein the balloon **72** extends over a tapered distal tip **84** and is tightly secured thereto to form a tight seal. Such an embodiment may provide ease of manufacture and a generally smooth transition from the distal tip to the balloon and to the prosthetic heart valve. The material of the balloon **72** may be secured to the material of the tapered distal tip **84** via various methods, such as adhesive and/or heat treatment.

[0061] FIG. 8 depicts a further embodiment of a distal end, wherein a proximal tapered piece **110** over which the balloon **72** passes is provided just proximal of the main body of the tapered distal tip **84**, with the balloon **72** at its distal end secured to the tapered tip **84** just distally of the proximal tapered piece **110** to form a tight seal. In the particular embodiment depicted, the balloon **72** is secured within a crevasse **112** formed between the proximal tapered piece and the tapered distal tip **84**. The prosthetic valve **86** is secured onto the balloon **72** proximally of the proximal tapered piece **110**. The proximal tapered piece **110** may preferably be formed of soft flexible material, such as the same material used for the tapered tip **84**. The tapered tip **84** and the proximal tapered piece **110** may be formed of a single piece of the same flexible/soft material. This assembly can provide a substantially continuous profile from the distal tip **82** to the crimped prosthetic valve **86**, thereby easing the passage of the prosthetic valve **86** into the native valve annulus.

[0062] FIGS. 9A-9B depict another embodiment of the invention, wherein a tapered distal tip **84** has a floppy and curled distal section **114**. The tapered distal tip **84** is formed from flexible material using known methods so that the distal section **114** is biased toward the curved configuration, as depicted in FIG. 9A. The curved configuration may be exceptionally atraumatic to tissue because the most distal portion **116** when curved will be presented as a "side" of the tapering material, as opposed to the distal tip **82** itself, and also because the curved portion **118** itself serves as a spring-like member to permit the curved tip to curve even more, and thus easily give way, when brought into contact with tissue. The curved portion **118** curves through an angle **122** of approximately 60

to 120 degrees, and in a more specific embodiment about 90 degrees or more, so that the distal tip **82** presents a distal tip surface which faces away from the longitudinal axis of the catheter **70** and/or faces at least partially proximally with respect to the catheter **70**, so that if the catheter **70** is advanced distally into tissue the distal tip **82** will contact the tissue sideways via a relatively smooth outer curved side **124** of the tapered section.

[0063] As depicted in FIG. **9B**, when the distal tip **82** of FIG. **9A** is positioned over a guidewire **120**, the guidewire **120** itself provides sufficient force to reconfigure the distal tip from the curved configuration into a substantially straight configuration. In the example depicted in FIGS. **9A** and **9B**, the tapered distal tip **84** tapers from a maximum width to a minimum width (at the distal tip **82**) along a length of 10 mm to 60 mm inches.

[0064] FIG. **10** depicts a tapered distal tip **84** similar to that of FIG. **9**, except the balloon **72** passes around the tapered distal tip and is secured thereto, similar to the fashion depicted previously with respect to FIG. **7**.

[0065] FIG. **11** depicts a proximal depicts a tapered distal tip **84** similar to that of FIGS. **9A-9B** and **10**, except that the tapered tip portion **84** includes a proximal portion **130** having a slightly smaller outer diameter than the middle portion **132**, so that a small recess **134** is created in proximal portion **130** into which a distal portion of the balloon **72** can be placed. The balloon **72** is secured to the proximal portion **130** via methods such as adhesive, heat treatment, etc., to create a tight seal which prevents fluid from leaking out of the balloon **72**. The resulting structure creates a relatively smooth outer surface at the transition **136** between the outer surface of the distal tip **84** and the balloon **72**. Note that the tapered distal tip **84** may be biased toward a curved configuration (similar to that depicted in FIG. **9A**), or the distal section may be biased toward a straightened configuration (such as that depicted in FIG. **9B**). A distal tip **84** according to the invention may comprise: as depicted in FIG. **11** (and also in FIGS. **9A, 9B, 10**), a concave surface **138** on the forward-facing portion of the distal tip; a convex surface (as depicted in FIG. **7**) on the forward-facing portion; or a flat angled surface (as depicted in FIG. **6**) on the forward-facing portion. The tapered surfaces (such as the concave surface of FIGS. **9A** to **11**, etc.) can facilitate advancement of the distal tip through the patient's anatomy, such as when the device is advanced (e.g., in retrograde direction) through a calcified heart valve (such as the aortic valve). For example, a forward-facing concave side surface (such as surface **138** of FIG. **11**) can engage against calcified/diseased valve leaflet edges and ease the leaflets aside to permit the catheter to be advanced through a calcified/diseased valve. Distal tips according to the invention may be formed from materials such as molded nylon, and may have maximum diameters of between 0.20 to 0.25 inches which taper to a tip diameter of 0.07 to 0.09 inches. The distal tips may have overall lengths of 0.6 to 0.7 inches, with (in an embodiment such as depicted in FIG. **11**) 0.1 to 0.2 inches of that length comprising the proximal portion **130**.

[0066] The delivery system described herein is particularly well-suited for a retrograde, transaortic approach, partly because of its relatively short length. With reference to FIG. **4**, the entire length of the introducer **50** is approximately 13 inches (33 cm), while the length of the sheath **52** that may extend within the body is about 8 inches. The portion of the balloon catheter **70** that extends into the introducer **50** (that is, the portion of the balloon catheter from the distal tapered tip **84** to approximately the handle **78**) is no more than about 24 inches (61 cm), which permits about 11 inches (28 cm) of the balloon catheter to extend beyond the introducer distal tip **72**. It should be noted that the relatively short length of the delivery system may be unsuited for a longer, more circuitous approach through the peripheral vasculature, such as shown in co-pending U.S. patent application Ser. No. 11/152,288. Also, the steering mechanism, if present, may be provided on the balloon catheter **70** itself, rather than on a secondary catheter used for guiding the balloon catheter, as is done in application Ser. No. 11/152,288. The short length of the balloon catheter and the ability to directly manipulate it greatly enhances successful positioning of the prosthetic heart in the aortic annulus.

[0067] While the invention has been described in its preferred embodiments, it is to be understood that the words which have been used are words of description and not of limitation. Therefore, changes may be made within the appended claims without departing from the true scope of the invention. The field of engineering design, which includes the design of structures, systems, devices, and components, is continuously seeking methods for improvement. The process of designing such items is complex and multifaceted, involving the consideration of numerous factors including functionality, efficiency, cost-effectiveness, safety, and regulatory compliance among others.

Claims

1. A surgical method of implanting a balloon-expandable prosthetic heart valve in a procedure while a patient's heart continues beating, the method comprising: creating a single puncture through an aortic wall at or near an aortic arch and inserting an introducer sheath having a lumen diameter no greater than 24 French; advancing, through the introducer sheath, a first catheter that carries a valvuloplasty balloon and includes a distal tapered tip that tapers, along 10 mm-40 mm, from (i) a proximal maximum outside diameter substantially equal to a valve-delivery diameter to (ii) a distal minimum outside diameter of 2 mm or less, the tip being formed of a polymer whose flexibility increases toward the distal end; inflating the valvuloplasty balloon to expand a stenotic aortic-valve annulus for pre-dilation, then deflating the balloon and withdrawing the first catheter without removing or replacing the introducer sheath; advancing, through the same sheath, a second catheter that carries the prosthetic heart valve in a radially crimped state and has a distal tapered tip formed within the same taper-length, maximum-diameter, minimum-diameter, and flexibility ranges as the first catheter; steering the second catheter so the prosthetic heart valve is positioned within the aortic annulus; inflating a delivery balloon of the second catheter to expand the prosthetic heart valve from the valve-delivery diameter to an expanded diameter in contact with the annulus; deflating and longitudinally elongating the delivery balloon; and withdrawing the second catheter and introducer sheath to complete the procedure.
2. The method of claim 1, wherein each tapered tip has a proximal outside diameter of 4 mm-10 mm and a distal outside diameter not greater than 1 mm.
3. The method of claim 1, wherein the distal tapered tip of each catheter comprises a polymer selected to provide sufficient flexibility to navigate through the left ventricular apex without causing trauma.
4. The method of claim 1, wherein each tapered tip encloses a guide-wire lumen sized to accept a 0.035-inch guide wire.
5. The method of claim 1, wherein the introducer sheath has a working length of 10 cm-40 cm and remains in place after insertion.
6. The method of claim 1, further comprising monitoring catheter advancement and valve deployment by fluoroscopy, echocardiography, or intravascular ultrasound.
7. The method of claim 1, wherein balloon inflation is performed while the balloon and prosthetic heart valve are continuously visualized with fluoroscopy or radiopaque contrast imaging.
8. The method of claim 1, further comprising injecting radiographic contrast medium through a lumen of the second catheter to verify absence of perivalvular leak.
9. The method of claim 1, further comprising leaving native aortic-valve leaflets in place such that the prosthetic heart valve in expanded configuration contacts native leaflets.
10. The method of claim 1, further comprising tightening at least one purse-string suture around the single puncture after the introducer sheath is withdrawn.
11. A method of implanting a balloon expandable prosthetic heart valve in a procedure while a patient's heart continues beating, the method comprising: introducing an introducer sheath through a wall of a left ventricle of a patient and advancing a distal end of the introducer sheath into the left

ventricle; advancing a pre-dilatation balloon catheter through the introducer sheath and through a native aortic valve; inflating a valvuloplasty balloon of the pre-dilatation balloon catheter to dilate the native aortic valve annulus; deflating and removing the pre-dilatation balloon catheter through the introducer sheath without removing or replacing the introducer sheath; advancing a valve-delivery catheter through the introducer sheath and into position to deploy a prosthetic heart valve at the native aortic valve; inflating a delivery balloon on the valve-delivery catheter to expand the prosthetic heart valve at the native aortic valve; and removing the valve-delivery catheter and the introducer sheath from the patient; wherein the pre-dilatation balloon catheter has a tapered distal tip formed in a substantially same fashion, and within a same taper-length range, as a distal tip of the valve-delivery catheter.

12. The method of claim 11, wherein the thoracic access incision is a right-anterior mini-thoracotomy at a second intercostal space.

13. The method of claim 11, further comprising placing first and second concentric purse-string sutures around the single puncture and cinching the sutures after the introducer sheath is removed.

14. The method of claim 11, wherein, during advancement of the second catheter, a deflection mechanism proximal to the delivery balloon is actuated to steer the prosthetic heart valve toward the annulus.

15. The method of claim 11, wherein a working length of each catheter inside the patient is no greater than 61 cm.

16. The method of claim 11, wherein the introducer sheath has an external hydrophilic coating.

17. The method of claim 11, wherein inflating the valvuloplasty balloon comprises introducing saline optionally mixed with radiopaque contrast to a pressure sufficient to enlarge the aortic annulus without bursting the balloon.

18. The method of claim 11, wherein the prosthetic heart valve comprises a balloon-expandable stent frame supporting flexible tissue leaflets.

19. The method of claim 11, wherein the delivery balloon is formed of nylon to increase rated burst pressure.

20. The method of claim 11, further comprising confirming, under fluoroscopy with contrast injection, that both coronary ostia remain unobstructed after prosthetic-valve deployment.

21. A method of sequentially dilating and valve-replacing a stenotic cardiac-valve annulus through a single introducer sheath, the method comprising: inserting an introducer sheath having a lumen diameter of 24 French or smaller through a vascular or cardiac access opening; advancing, through the sheath, a first catheter whose distal tapered tip tapers, along 10 mm-40 mm, from a proximal maximum outside diameter substantially equal to a valve-delivery diameter to a distal minimum outside diameter of 2 mm or less, the tip being progressively more flexible toward its distal end, the first catheter carrying a valvuloplasty balloon; inflating the valvuloplasty balloon to enlarge the valve annulus, then deflating the balloon and withdrawing the first catheter while leaving the introducer sheath in place; advancing, through the same introducer sheath, a second catheter that carries a balloon-expandable prosthetic heart valve and whose distal tapered tip conforms to the same taper-length, maximum-diameter, and minimum-diameter ranges of the first catheter; positioning the prosthetic heart valve within the valve annulus and inflating a balloon of the second catheter to expand the prosthetic heart valve into operative engagement with the annulus; and deflating the balloon and removing the second catheter and introducer sheath.

22. The method of claim 21, wherein the access opening is formed in an aortic arch.

23. The method of claim 21, wherein the access opening is formed in a left-ventricular apex.

24. The method of claim 21, wherein, during expansion of the balloon-expandable prosthetic heart valve, a physician maintains real-time fluoroscopic visualization to verify valve position relative to the aortic annulus.

25. The method of claim 21, wherein each catheter's distal tapered tip includes a self-biased curved portion that straightens when advanced over a guide wire.

26. The method of claim 21, further comprising advancing a 0.035-inch guide wire through the sheath and across the valve annulus.

27. The method of claim 21, further comprising injecting contrast medium through a lumen of the second catheter and visualizing flow to assess valve seating.

28. The method of claim 21, wherein the introducer sheath has an overall length no greater than 33 cm.

29. The method of claim 21, further comprising tightening at least one purse-string suture around the access opening after the introducer sheath is removed.

30. The method of claim 21, wherein the prosthetic heart valve is expanded while resident native valve leaflets remain in place within the annulus.
